BASIC HTML & CSS

A concise reference for the tags, properties, and layout concepts beginners use most.

How to use this guide: Read the short notes, then try each property in a small HTML file. CSS becomes easier when you change one value at a time and observe the result.

HTML gives content structure. CSS controls how that content looks and is arranged.

1. HTML: Essential Tags

HTML elements describe the meaning and structure of page content. Most have an opening tag, content, and a closing tag.

Tag	What it does	Example
<!DOCTYPE html>	Tells the browser to use modern HTML.	At the top of an HTML document.
<html>	Wraps the whole HTML document.	The root element.
<head>	Holds page information and linked resources.	Contains title, metadata, CSS links.
<title>	Sets the browser-tab title.	<title>My Page</title>
<body>	Contains visible page content.	Headings, text, images, buttons.
<h1>–<h6>	Headings, from most important to least important.	Welcome
	A paragraph of text.	Hello
<a>	Creates a link; use href for its destination.	About
<img>	Displays an image; use src and useful alt text.	<img src="photo.jpg" alt="Portrait">
	Generic block container for grouping content.	...
	Generic inline container, often for styling part of text.	new
/	Unordered (bullets) or ordered (numbered) list.	Item
	A list item inside a list.	First
<button>	A clickable button.	<button>Save</button>
<input>	Accepts user input; its type controls the kind.	<input type="email">
<form>	Groups controls for submitting user data.	Contains inputs and a submit button.
<label>	Names a form control; connect it using for/id.	<label for="email">Email</label>
<header>	Introductory content for a page or section.	Often contains a logo or heading.
<nav>	A group of navigation links.	Main site menu.
<main>	The page's primary content.	Usually one per page.
<section>	Groups related content, usually with a heading.	A features section.
<article>	Self-contained content that can stand alone.	A blog post or news item.
<footer>	Footer information for a page or section.	Copyright or contact details.
/	Marks strong importance / emphasis.	Usually appears bold / italic.

A minimal HTML document

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Page</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Hello, world!</h1>
```

```
<p>My first web page.</p>
</body>
</html>
```

The viewport line helps the page scale properly on mobile. The link element connects an external CSS file.

2. CSS: The Basics

CSS rules select HTML elements and apply declarations. A declaration is a property and its value.

```
selector {
property: value;
}

p {
color: navy;
font-size: 16px;
}
```

Concept	Meaning
Selector	Chooses which elements to style, such as p, .card, or #header.
Property	The aspect you want to change, such as color or padding.
Value	The setting given to a property, such as blue or 12px.
Class selector (.name)	Styles every element with that class. Classes can be reused.
ID selector (#name)	Targets an element with that ID. IDs should be unique on a page.
Element selector	Targets every element of a type, e.g. h1 or p.
Grouping (h1, p)	Applies the same declarations to multiple selectors.
Comments	Notes ignored by the browser: /* comment */.

Ways to add CSS

Method	Example	When to use
External	<link rel="stylesheet" href="style.css">	Best default for most projects; keeps styles separate.
Internal	<style> p { color: red; } </style>	Useful for a small one-page example.
Inline	<p style="color:red">	Avoid for normal styling; harder to maintain.

3. The Box Model: Margin, Border, Padding

Every element is treated like a box. From the inside out: content → padding → border → margin.

Property	What it means	Remember
content	The text, image, or other content inside the element.	The inner area.
padding	Space between the content and its border.	Adds breathing room inside the box.
border	The line surrounding padding and content.	Can have width, style, and color.

Property	What it means	Remember
margin	Space outside the border, separating the element from others.	Creates distance between boxes.

Quick comparison: padding makes a box feel roomier inside; margin pushes other elements away. Background colors usually extend through content and padding, but not margin.

```
.card {
padding: 20px;
margin: 16px;
border: 1px solid #ccc;
}
```

Useful box-sizing rule

```
*,
*::before,
*::after {
box-sizing: border-box;
}
```

With border-box, the declared width includes padding and border. This makes sizing easier to predict.

Spacing shorthand

Code	Meaning
margin: 10px;	All four sides.
margin: 10px 20px;	Top/bottom 10px; left/right 20px.
margin: 10px 20px 30px;	Top 10px; left/right 20px; bottom 30px.
margin: 1px 2px 3px 4px;	Top, right, bottom, left (clockwise).
padding-inline: 16px;	Left and right in common horizontal writing.
padding-block: 12px;	Top and bottom in common horizontal writing.

4. Sizing, Display, and Position

Property	What to know
width / height	Sets an element's size. Fixed sizes can be awkward on small screens.
max-width	Caps how wide an element can grow; useful for readable content containers.
min-height	Sets a minimum height while allowing the element to grow.
display: block	Starts on a new line and generally takes available width.
display: inline	Flows with text; width and height usually do not apply as expected.
display: inline-block	Flows inline but accepts width and height.
display: none	Removes the element from layout and hides it.
position: static	Default positioning in normal document flow.
position: relative	Stays in flow; can be offset and can anchor absolutely positioned children.
position: absolute	Removed from normal flow and positioned relative to a suitable positioned ancestor.

Property	What to know
position: fixed	Usually positioned relative to the viewport and stays there while scrolling.
position: sticky	Acts normally until a scroll threshold, then sticks within its scrolling area.
top/right/bottom/left	Offsets used with positioned elements.
overflow	Controls content that exceeds a box: visible, hidden, auto, or scroll.
z-index	Controls stacking order in applicable stacking contexts; higher is not always globally on top.

A simple centered container

```
.container {
width: 90%;
max-width: 1100px;
margin-inline: auto;
}
```

The automatic horizontal margins center a block when there is available space and its width is constrained.

5. Flexbox: One-Dimensional Layout

Flexbox arranges items in one main direction: a row or a column. Apply display: flex to the parent (flex container), then control the children (flex items).

```
.container {
display: flex;
flex-direction: row;
justify-content: space-between;
align-items: center;
gap: 16px;
}
```

Property	Applies to	Meaning
display: flex	Parent	Turns children into flex items.
flex-direction	Parent	row, column, or reverse variants; sets the main axis.
justify-content	Parent	Distributes items along the main axis: center, space-between, space-around, etc.
align-items	Parent	Aligns items across the cross axis: stretch, center, flex-start, etc.
gap	Parent	Adds space between items without adding space around the outside.
flex-wrap	Parent	wrap lets items move onto another line if needed.
align-content	Parent	Distributes multiple flex lines when wrapping creates extra cross-axis space.
flex	Child	Shorthand controlling how an item grows, shrinks, and its base size.
flex-grow	Child	How much an item can grow when extra space exists.

Property	Applies to	Meaning
flex-shrink	Child	How much an item can shrink when space is limited.
flex-basis	Child	The initial main-size basis before free space is distributed.
align-self	Child	Overrides the container's align-items for one item.
order	Child	Changes visual order; avoid if it makes the reading/tab order confusing.

Axis tip: with flex-direction: row, the main axis is horizontal and the cross axis is vertical. With column, those axes switch.

Common Flexbox patterns

```
/* Center content in both directions */
.center {
display: flex;
justify-content: center;
align-items: center;
}

/* Navigation row with spacing */
.nav {
display: flex;
align-items: center;
gap: 20px;
}

/* Allow cards to wrap */
.cards {
display: flex;
flex-wrap: wrap;
gap: 16px;
}
```

6. CSS Grid: Rows and Columns

Grid is useful for two-dimensional layouts where you control rows and columns together. Flexbox is often simpler for a single row or column.

```
.grid {
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 16px;
}
```

Property	Meaning
display: grid	Turns the parent into a grid container.
grid-template-columns	Defines columns, e.g. 1fr 2fr or repeat(3, 1fr).
grid-template-rows	Defines row sizes.
gap	Space between grid rows and columns.
fr	A fraction of the available grid space.
grid-column	Controls which columns an item spans or occupies.
place-items	Shorthand for aligning items on both axes in many grid cases.

Responsive grid

```
.grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
gap: 16px;
}
```

This creates as many columns as fit, with each column aiming to be at least 220px wide and sharing remaining space.

7. Typography, Color, and Backgrounds

Property	What it controls / useful values
color	Text color, e.g. #222, navy, rgb(...).
background-color	Background fill behind an element.
background-image	Adds an image or gradient behind content.
font-family	Typeface; provide fallback fonts, e.g. Arial, sans-serif.
font-size	Text size; rem is often useful for scalable type.
font-weight	Thickness, commonly 400 normal and 700 bold.
font-style	Normal or italic text.
line-height	Space between lines; unitless values like 1.5 are often convenient.
text-align	Aligns inline text: left, center, right.
text-decoration	Underline, line-through, or none.
text-transform	Changes visual capitalization: uppercase, lowercase, capitalize.
letter-spacing	Space between letters.
border-radius	Rounds corners; e.g. 8px or 50%.
box-shadow	Adds a shadow around a box.
opacity	Transparency from 0 (invisible) to 1 (opaque).

Color formats

Format	Example
Named color	tomato
Hex	#2563eb or #fff
RGB / RGBA	rgb(37 99 235) / rgba(0, 0, 0, 0.5)
HSL	hsl(221 83% 53%)

8. Units and Responsive Design

Unit / feature	Meaning
px	Pixels; a fixed CSS length unit.
%	Relative to a relevant containing size; exact reference depends on the property.
em	Relative to the element's font size in many common uses.

Unit / feature	Meaning
rem	Relative to the root element's font size; useful for consistent type and spacing.
vw / vh	A percentage of viewport width / height.
fr	Fractional unit used in CSS Grid tracks.
min(), max(), clamp()	Functions for choosing responsive values. Example: font-size: clamp(1.5rem, 4vw, 3rem).
Media query	Applies styles only when a condition is true, such as a narrow screen.

```
/* Base styles first */
.card-grid {
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 16px;
}

/* Adjust for smaller screens */
@media (max-width: 700px) {
.card-grid {
grid-template-columns: 1fr;
}
}
```

Responsive design means the layout adapts to different screen sizes. Prefer flexible widths, max-width, wrapping, and media queries over assuming every screen is the same.

9. Selectors, States, and Specificity

Selector / concept	Meaning
.card p	Descendant selector: paragraphs anywhere inside an element with class card.
.card > p	Child selector: only direct paragraph children.
h1, h2	Group selector: both heading types.
a:hover	Pseudo-class: style a link while the pointer hovers over it.
input:focus	Styles a focused input, including keyboard focus.
:first-child	Selects an element when it is the first child of its parent.
::before / ::after	Pseudo-elements for generated decorative content; set content.
Specificity	When declarations conflict, more specific selectors generally win; source order breaks ties. Cascade layers, importance, and inheritance also matter.
Inheritance	Some properties, such as color and font-family, often pass from parent to child; others, such as margin, do not.

```
.button:hover {
background-color: #1d4ed8;
}

input:focus {
outline: 2px solid #2563eb;
outline-offset: 2px;
}
```

Keep focus indicators visible for keyboard users. Do not rely on hover alone to communicate important information.

10. Images, Lists, and Common UI Details

Property	Use
object-fit	Controls how an image/video fits its box: cover fills and may crop; contain shows the whole item.
object-position	Controls which part of an image is visible when cropped.
list-style	Controls list markers and their appearance.
cursor	Controls the pointer cursor, e.g. pointer for clickable controls.
transition	Animates property changes smoothly, e.g. color or transform.
transform	Moves, rotates, scales, or skews an element without changing normal flow.
outline	Draws an outline outside the border; useful for focus states.

```
.thumbnail {
width: 100%;
aspect-ratio: 16 / 9;
object-fit: cover;
border-radius: 8px;
}

.button {
transition: background-color 180ms ease;
}
```

11. Handy CSS Properties: Quick Checklist

Category	Properties to recognize
Spacing	margin, padding, gap
Sizing	width, height, min-width, max-width, min-height, max-height
Layout	display, position, top, right, bottom, left, overflow, z-index
Flexbox	flex-direction, justify-content, align-items, flex-wrap, flex
Grid	grid-template-columns, grid-template-rows, grid-column, gap
Text	font-family, font-size, font-weight, line-height, text-align
Appearance	color, background, border, border-radius, box-shadow, opacity
Responsive	max-width, flex-wrap, minmax(), clamp(), @media

12. A Small Practice Example

Create two files named index.html and style.css in the same folder. This example combines the main ideas: spacing, typography, a centered container, Flexbox, and responsive sizing.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Profile Card</title>
```

```
<link rel="stylesheet" href="style.css">
</head>
<body>
<main class="container">
<section class="card">
<h1>My Profile</h1>
<p>I am learning HTML and CSS.</p>
<a class="button" href="#about">Learn more</a>
</section>
</main>
</body>
</html>
```

style.css

```
*,
*::before,
*::after {
box-sizing: border-box;
}

body {
margin: 0;
font-family: Arial, sans-serif;
line-height: 1.5;
color: #172033;
background-color: #f1f5f9;
}

.container {
width: 90%;
max-width: 900px;
margin: 40px auto;
}

.card {
padding: 24px;
border: 1px solid #cbd5e1;
border-radius: 12px;
background-color: white;
}

.button {
display: inline-block;
margin-top: 12px;
padding: 10px 16px;
border-radius: 8px;
color: white;
background-color: #1d4ed8;
text-decoration: none;
}

.button:hover {
background-color: #1e40af;
}

@media (max-width: 600px) {
.container {
width: 94%;
margin-top: 20px;
}

.card {
padding: 16px;
}
}
```

Try changing the card's padding and margin separately. Then add a second card and use Flexbox or Grid to arrange both. Observe what changes on a narrow screen.

13. Beginner Reminders

• HTML describes content; CSS styles and lays it out.

• Use classes for reusable styling and meaningful semantic HTML where possible.

• Padding is inside the border; margin is outside the border; gap is spacing between layout items.

• Flexbox is usually best for one-dimensional layouts; Grid is useful for rows and columns together.

• Use the browser's developer tools to inspect elements and experiment with CSS.

• Build mobile-friendly layouts early, and test with keyboard navigation as well as a mouse.

• CSS properties can interact. If a change does not appear, check the selector, cascade, specificity, and browser developer tools.